

Digestive System

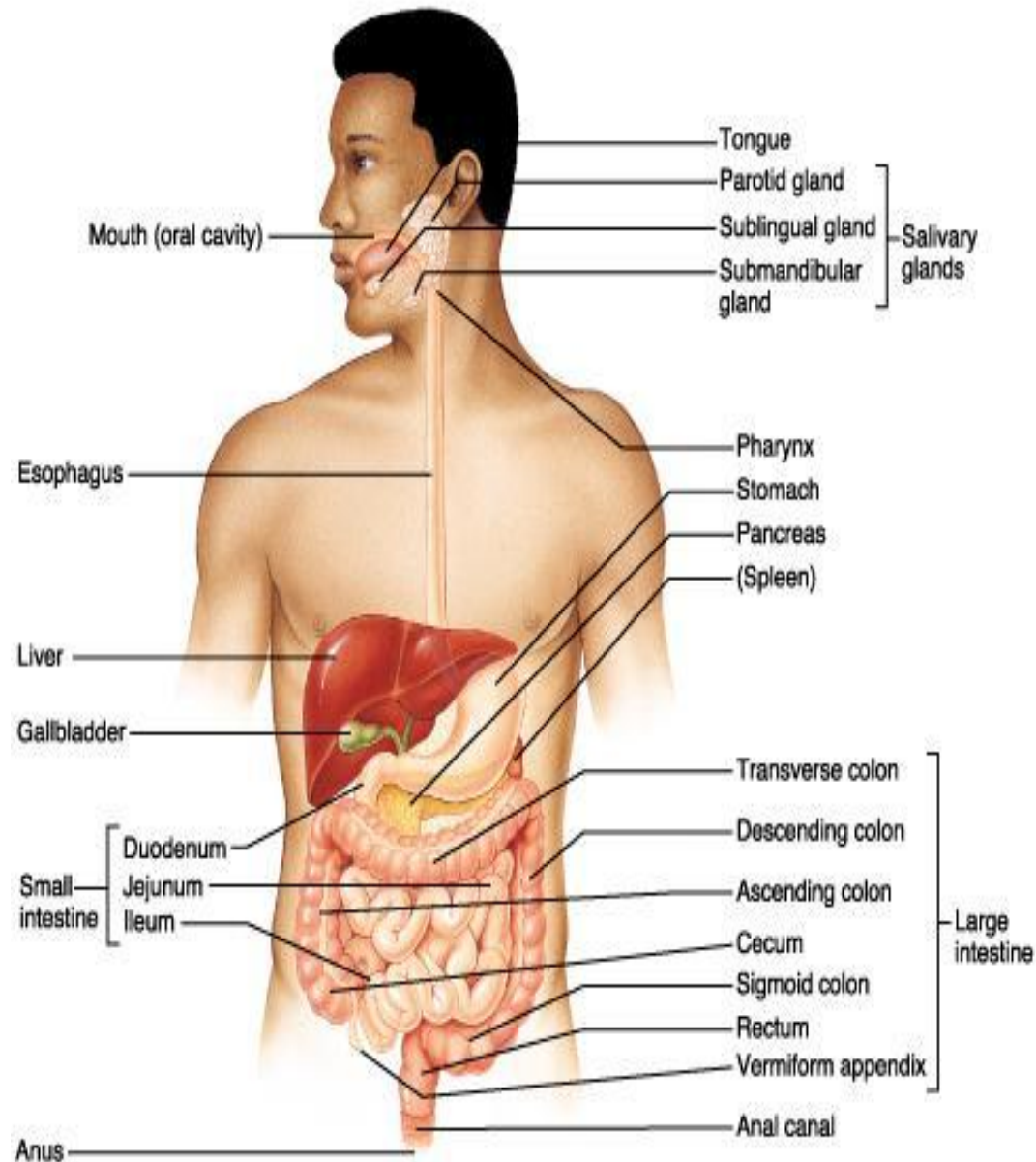
Part (1)

Digestive System

♣ Consists of digestive tract and associated glands

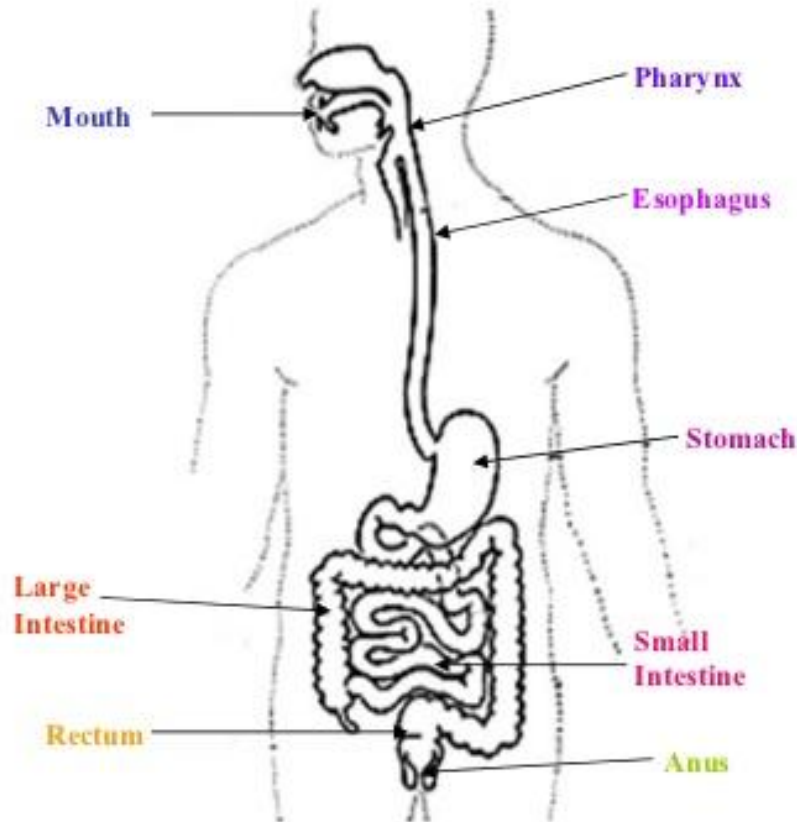
♣ **The digestive tract consists** of mouth, pharynx, esophagus, stomach, small intestine and large intestine

♣ **The digestive glands** include; salivary glands, liver, gall bladder and pancreas



The **digestive system** is a collection of organs that can be divided into two parts:

1. The **alimentary** or **gastrointestinal tract**.



2. The **accessory organs**



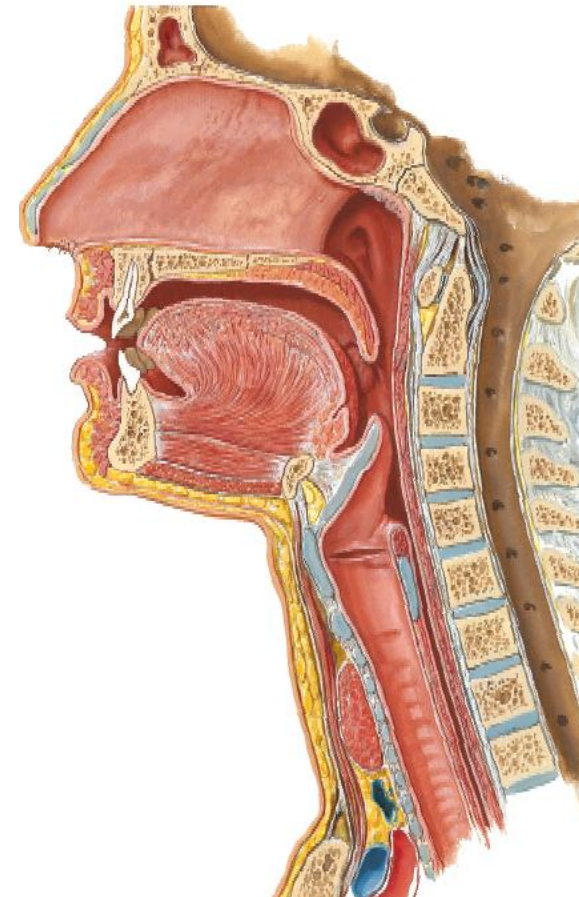
Alimentary Canal

- Alimentary canal (gastrointestinal tract) is a continuous coiled hollow muscular tube which occupies the ventral body cavity and is opened at both ends.
- The organs or parts of alimentary canal are:
- mouth., pharynx., esophagus, stomach, small intestine, large intestine, anas.

1- Mouth (oral cavity)

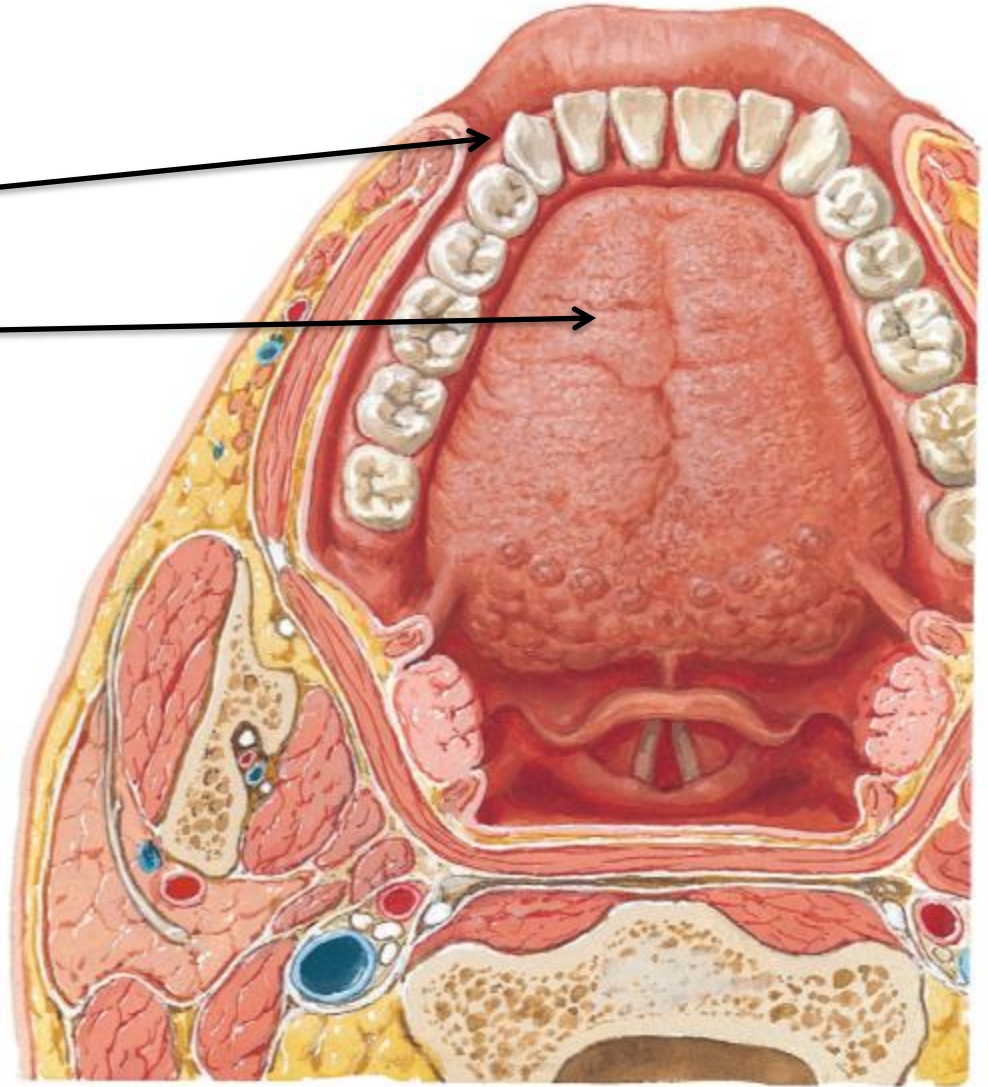
- Is a mucus membrane lined cavity . Lips protect its anterior opening,.
- The cheeks form its lateral walls.
- The hard palate forms its anterior roof, the soft palate forms its posterior roof,.
- Vestibule of the mouth is the cavity lying between the cheeks and lips externally and the teeth and gums internally.
- The oral cavity proper lies within the teeth and alveolar process of maxilla and mandible which communicates with the vestibule behind the third molar teeth,.
- The tongue is attached to the floor of the mouth by the frenulum.

Pharynx: Median Section



1-Mouth & Oral Cavity

- ♣ The oral cavity is divided into small vestibule and larger mouth cavity proper

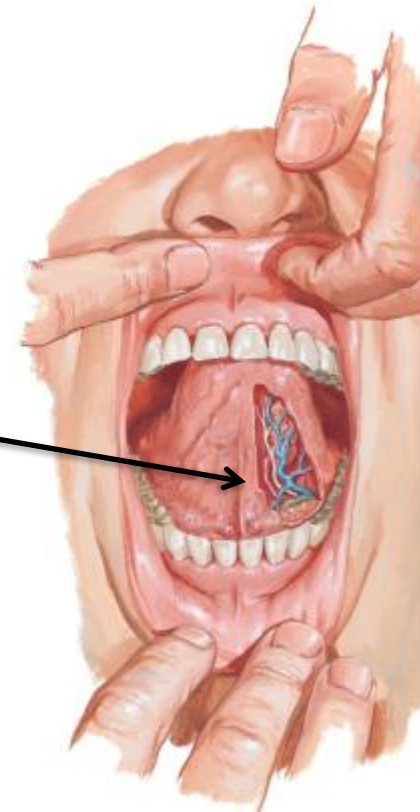


♣ The mouth cavity proper has Roof and Floor

Roof :
Hard and soft palate (Uvula)

Floor:
tongue (Frenulum of tongue)

contains tongue, teeth, sublingual glands and openings of salivary ducts

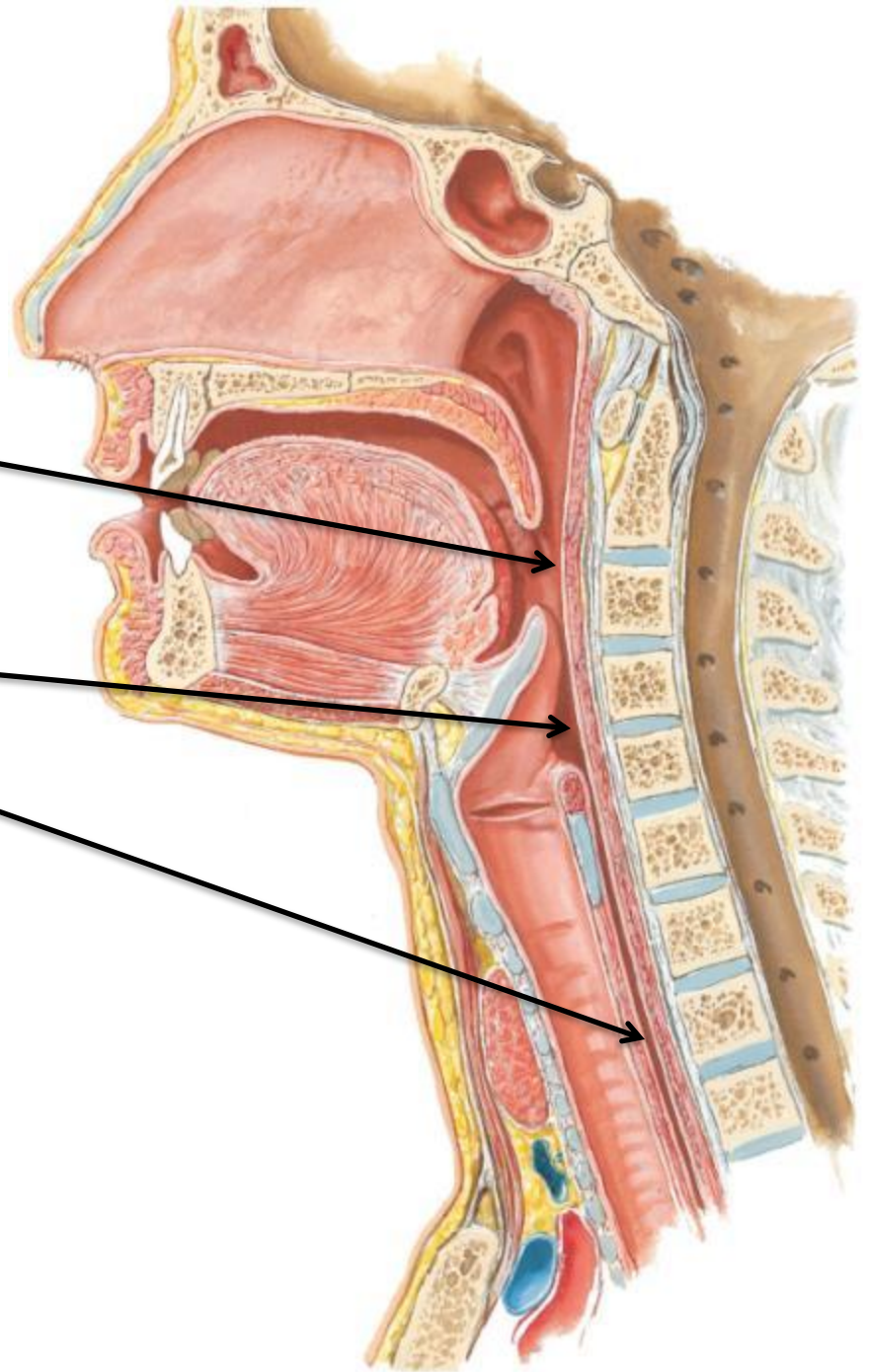


2- The Pharynx

The oral cavity ends into the **oropharynx**

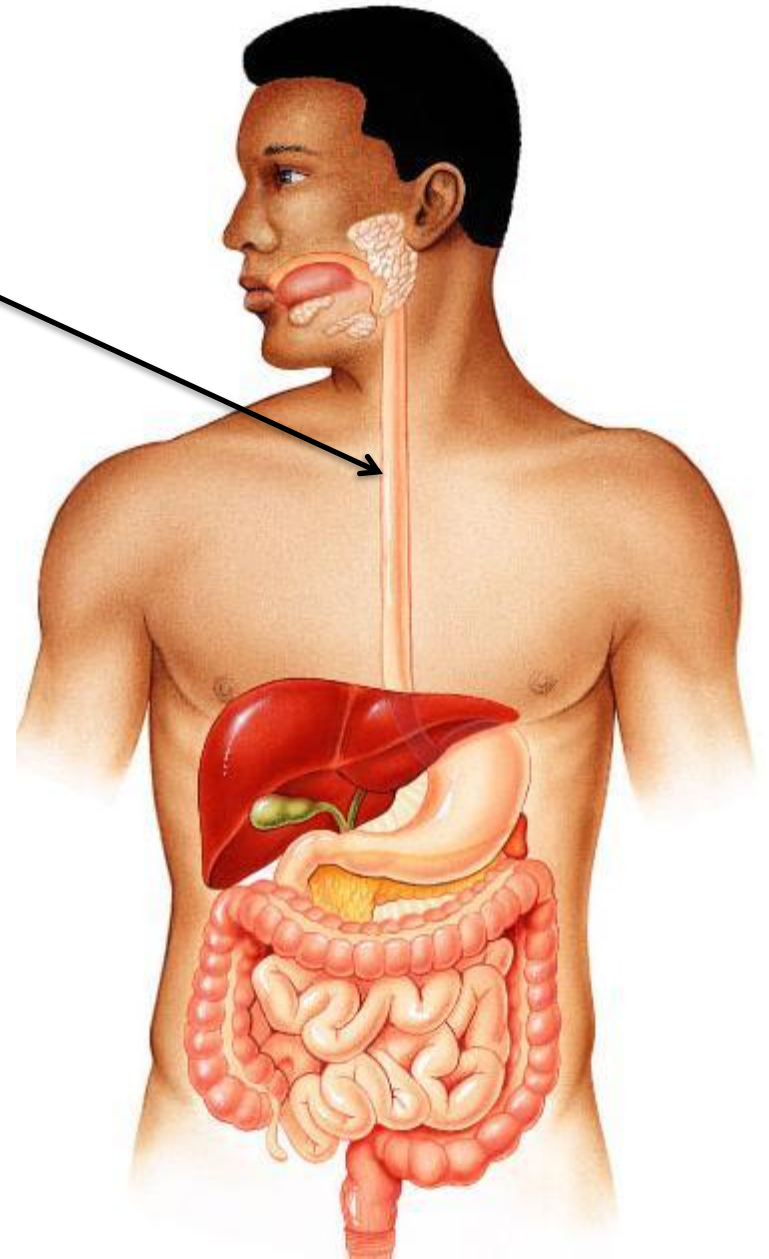
The oropharynx continues as **laryngopharynx**

which leads to **esophagus**

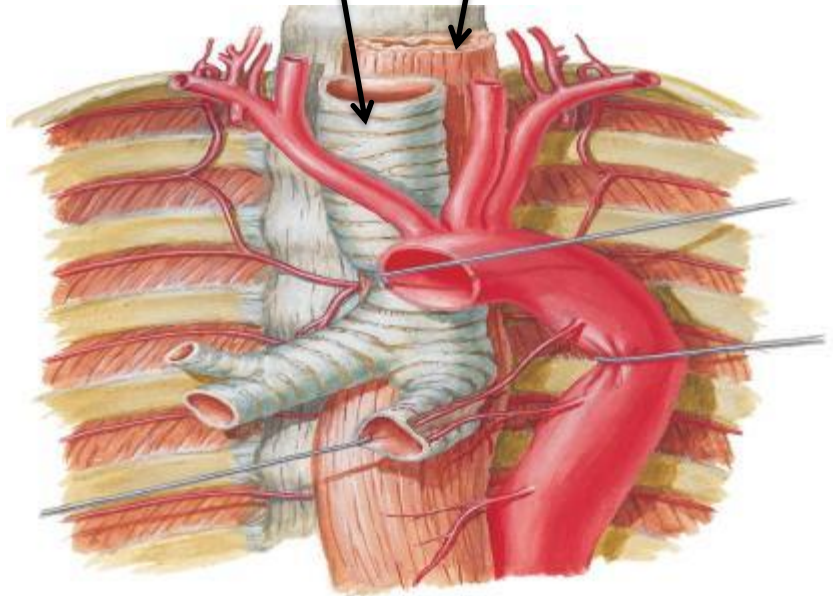
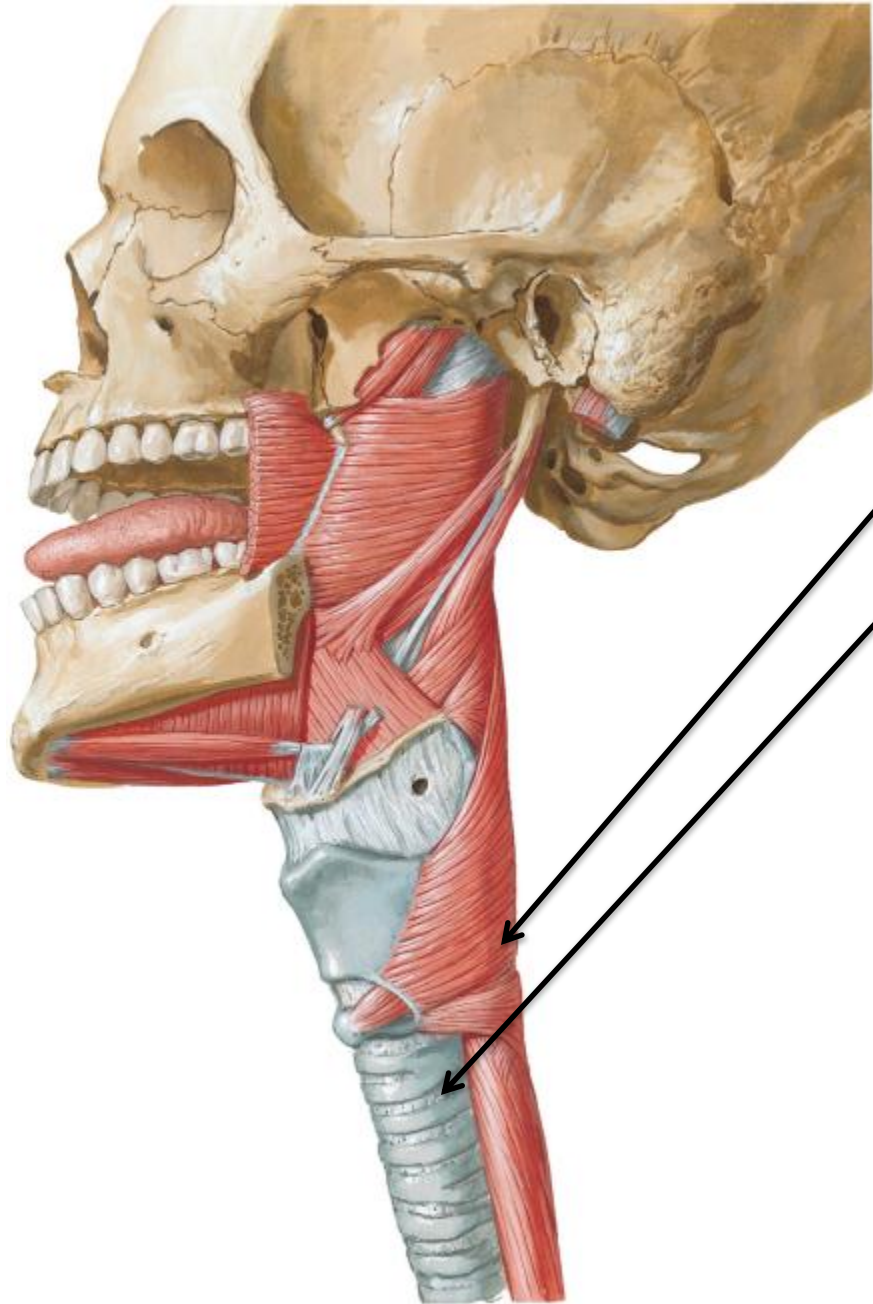


3-Esophagus

- ♣ 25 cm long muscular tube begins from the end of laryngo-pharynx and ends in the cardiac opening of stomach
- ♣ It has cervical, thoracic and short abdominal parts
- ♣ It delivers the ingested food from pharynx to stomach by peristalsis movement



Relation of Esophagus & trachea



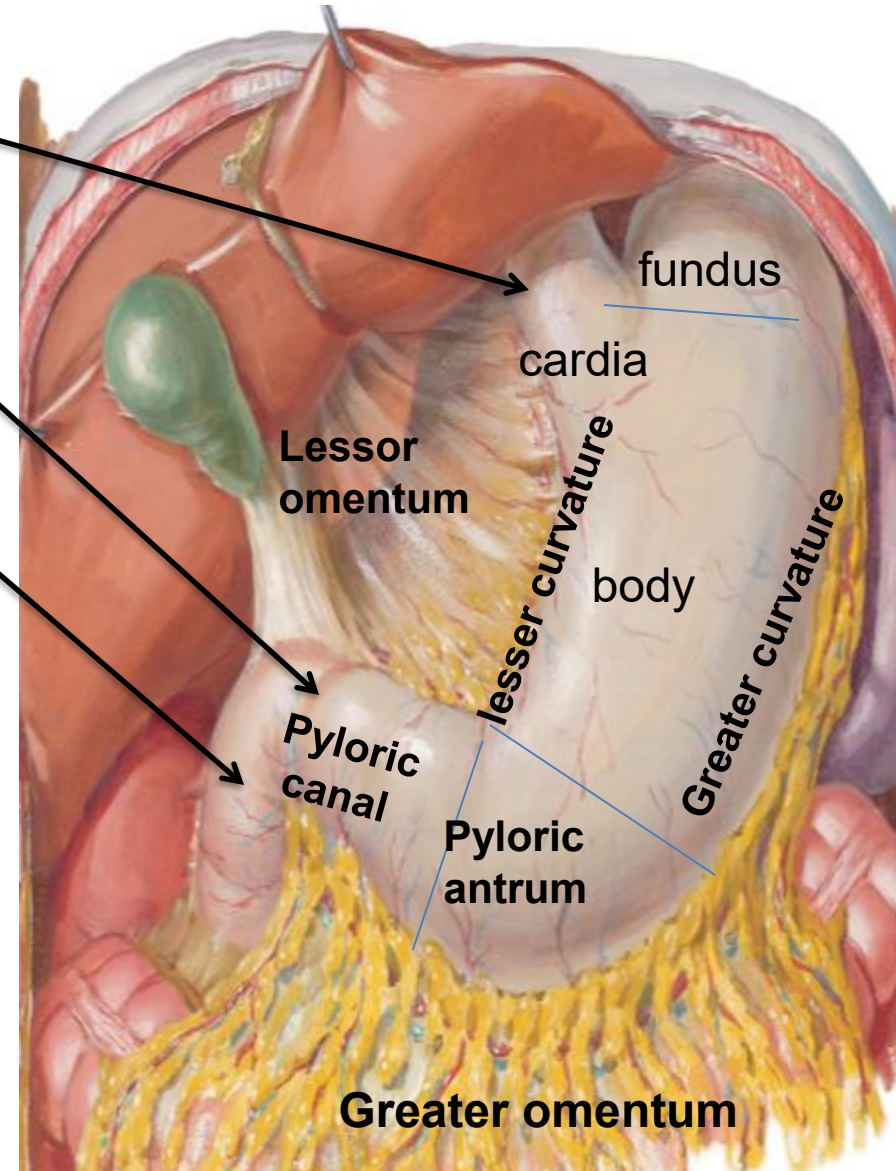
4- The Stomach

♣ Begins at the cardiac opening and ends at the pyloric opening (junction) with duodenum

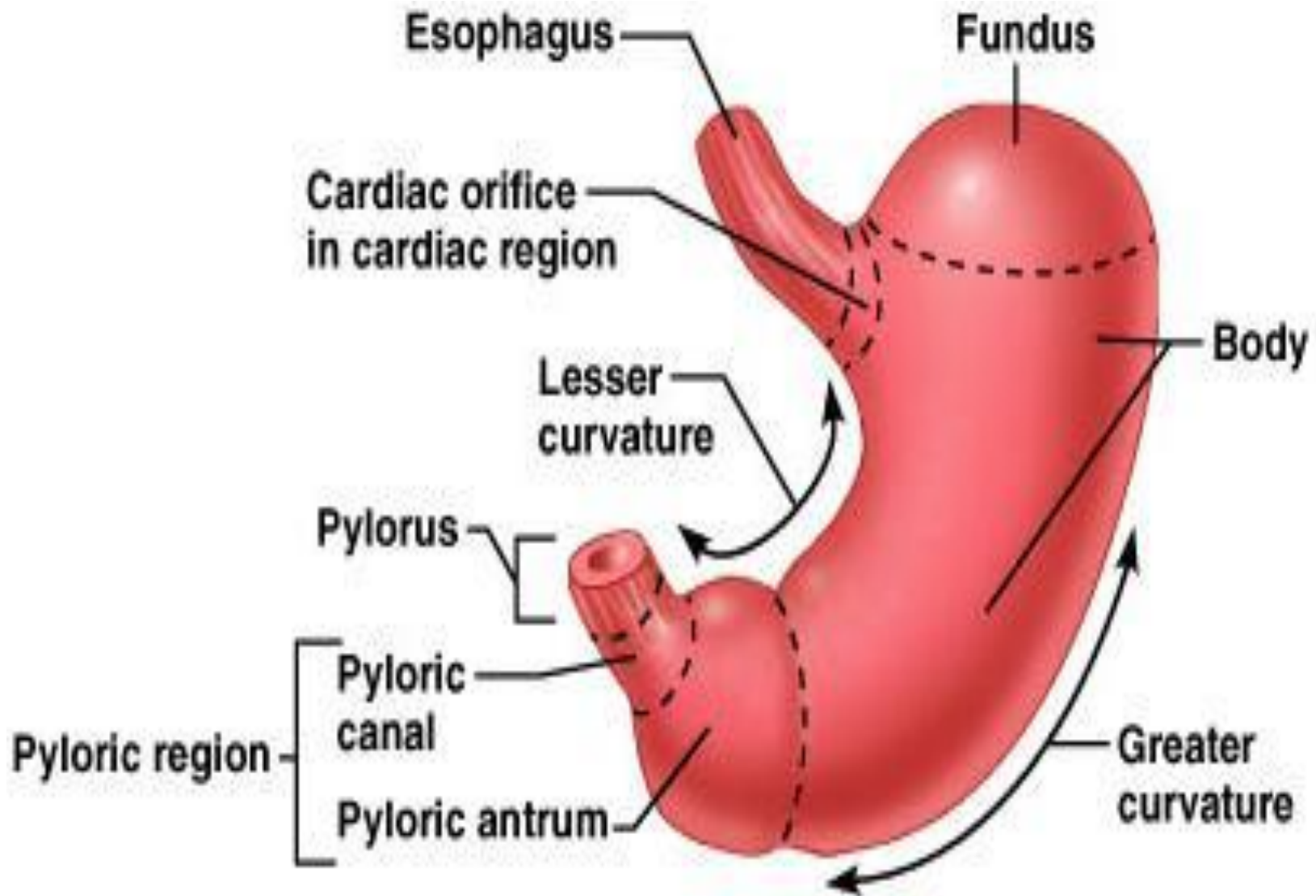
♣ Has two borders; greater curvature to the left and lesser curvature to the right

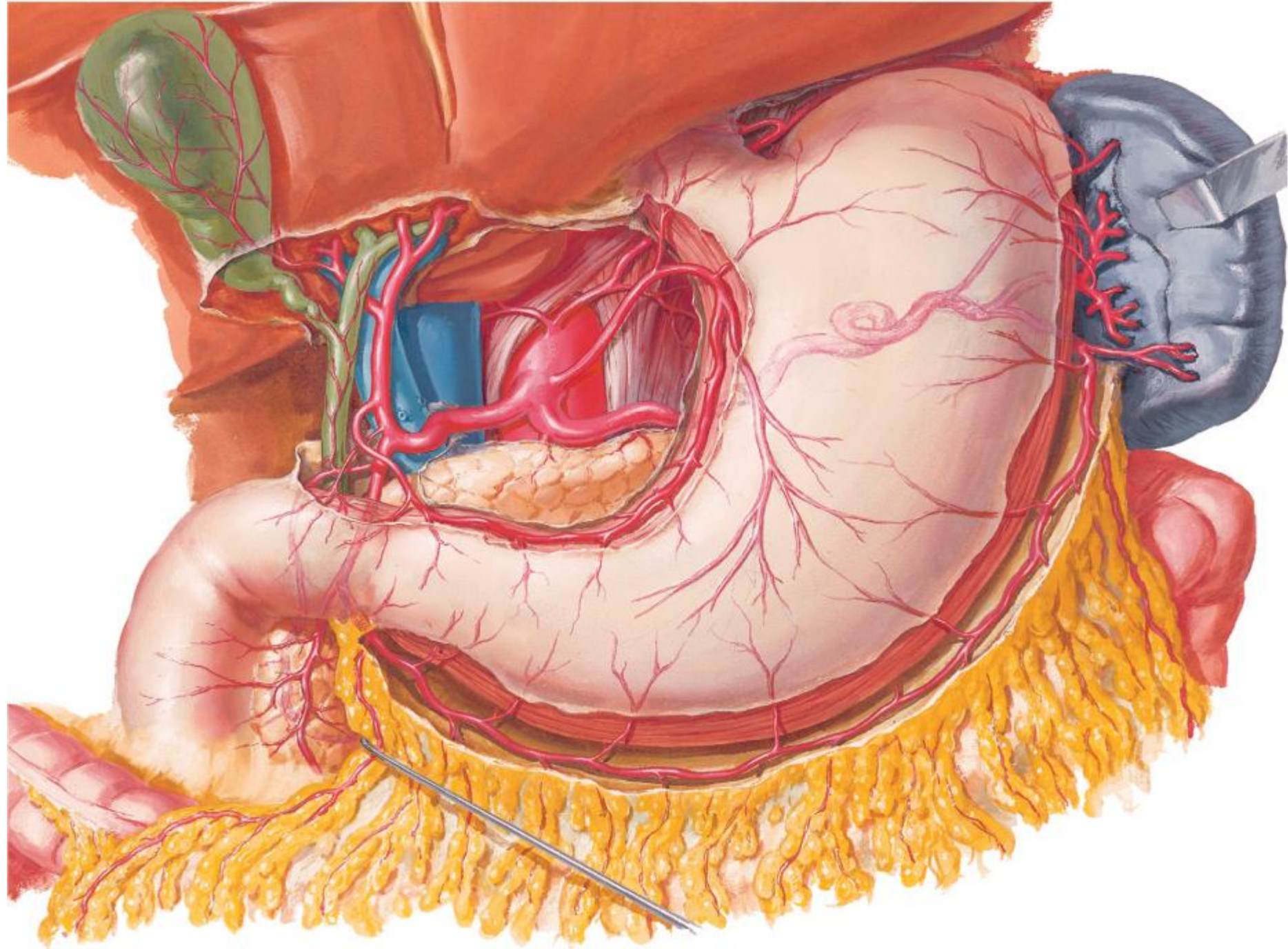
♣ It is descriptively divided into; cardia, fundus, body, antrum & pyloric canal

Lesser omentum is attached to lesser curvature. **Greater omentum** is attached to greater curvature.



4- Stomach





Small intestine

- Is the longest portion of alimentary canal (6 meters)
- Is of three parts: duodenum, jejunum, ileum.
- It starts at pylorus and ends at ileocecal junction.
- Duodenum is 25 cm long, c shaped and surrounds the head of pancreas. It receives the opening of bile duct and major and minor pancreatic ducts.
- Most of absorption of food happens in small intestine.
- The surface area of small intestine is increased by presence of microvilli on the cell membrane of its epithelial cells.
- Also villi which are fingerlike projections of the mucosa.
- Plicae circularis are large circular folds of mucosa and submucosa.
- Peyer's patches are lymphoid nodules in the submucosa of distal parts of small intestine

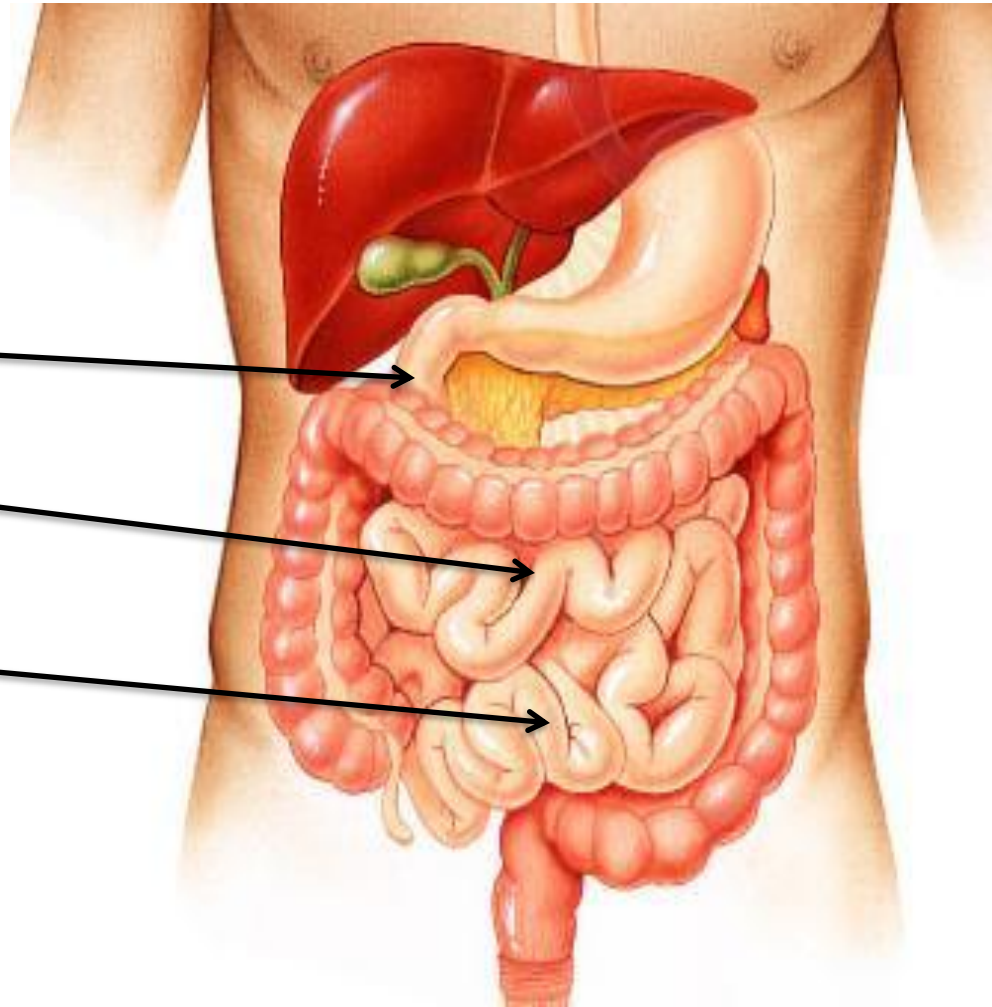
5- The Small Intestine

5-6 m long, consists of

Duodenum

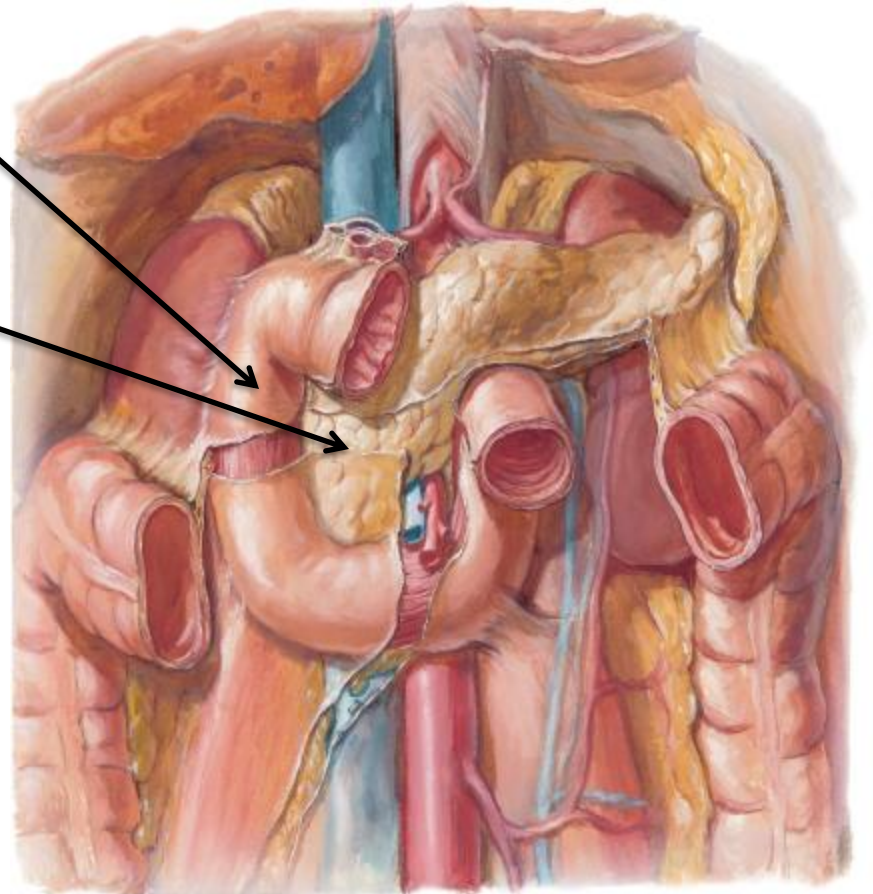
Jejunum and

Ileum



A-Duodenum

- ♣ is 12 inches long, C-shaped and surround the pancreas
- ♣ It has the openings of bile and pancreatic ducts
- ♣ it digests the food particles by bile from liver, and pancreatic



B- Jejunum

Coiled muscular tube, 1.5 m long

C- Ileum

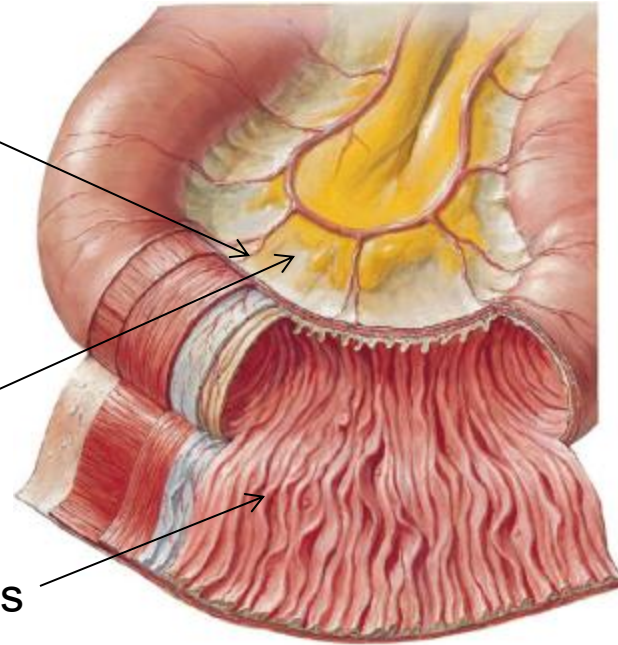
♣ Coiled muscular tube, 3.5 m long

♣ *Its function is absorption of digested food particles*

Anastomotic loops (arcades) of jejunum arteries(1 or 2)

Fat near the root

Circular folds

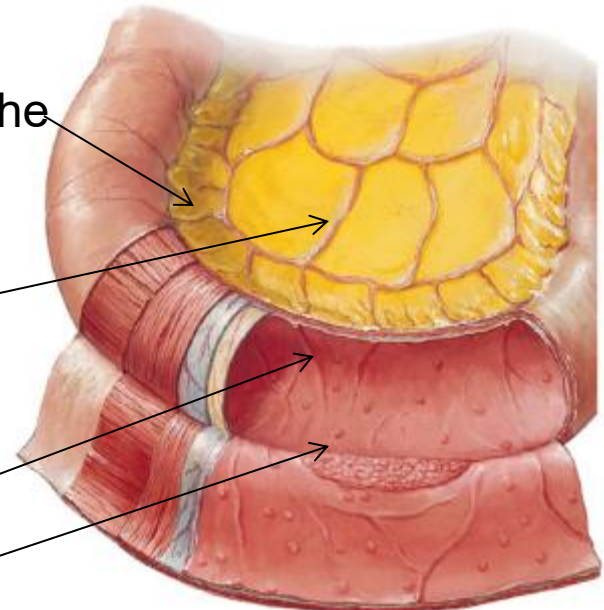


Fat over the root

Anastomotic loops (arcades) of ileal arteries(3 or 4)

Circular folds

Aggregate lymphoid nodules (Peyer's patches)



Large intestine

- Larger in diameter than small intestine but much shorter.
- It extends from the ileocecal junction to the anal canal

Parts of the L.N:

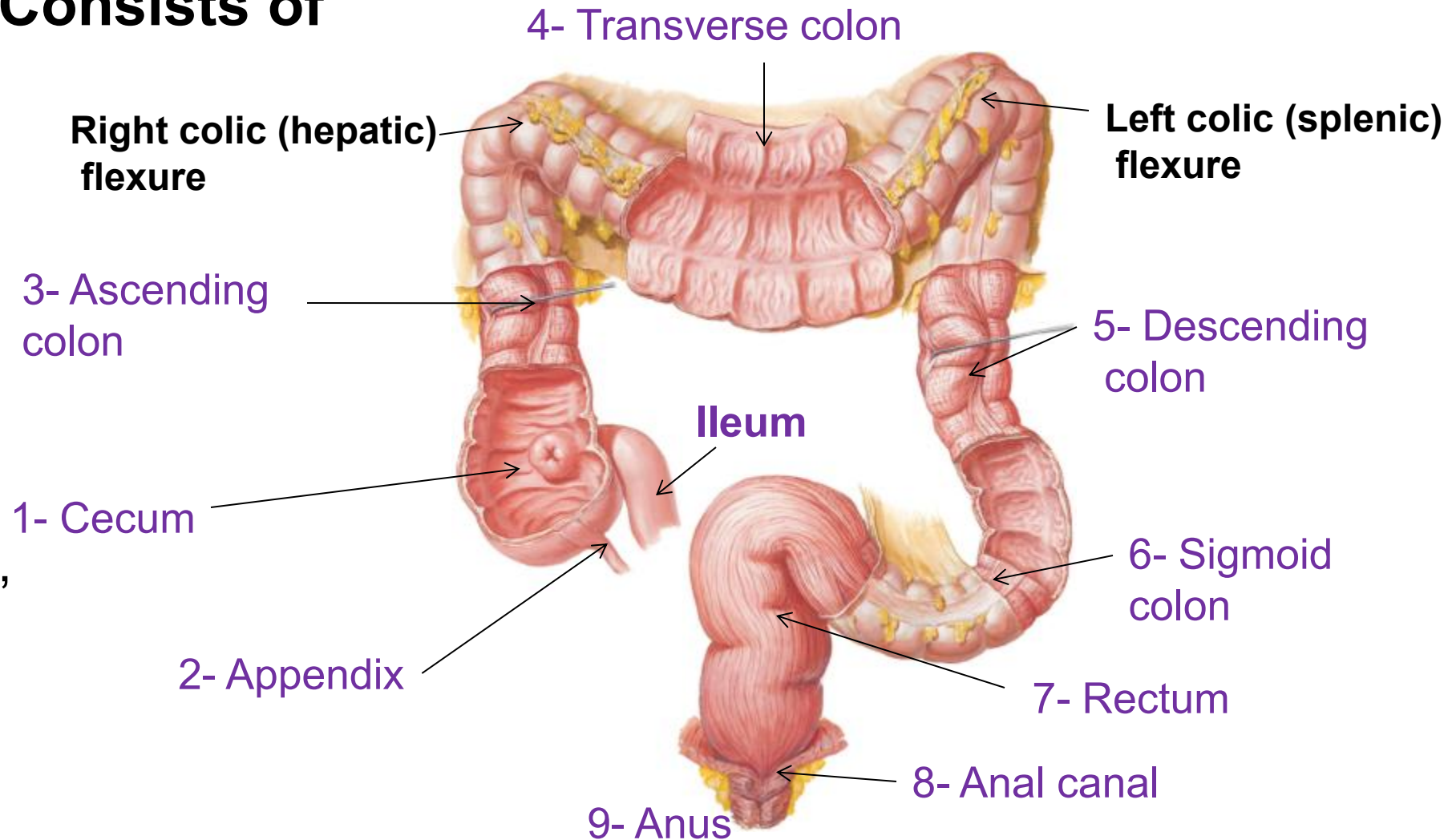
- Cecum
- Ascending colon
- Descending colon
- Rectum
- Appendix
- Transverse colon
- Sigmoid colon
- Anal canal

- The main function of large intestine is to absorb water and electrolytes and to store the residues of digestion and its elimination from the body as feces.
- There are no villi in the large intestine
- Appendix is commonly inflamed and infected (appendicitis) and then appendectomy may be necessary.
- The anal canal has external voluntary sphincter and internal involuntary anal sphincter.
- The external sphincter is skeletal muscle while the internal sphincter is smooth muscle
- Three bands of smooth muscle are present in the outermost layer of colon called tenia coli.
- The wall of large intestine is sacculated called haustra.

6- The Large Intestine

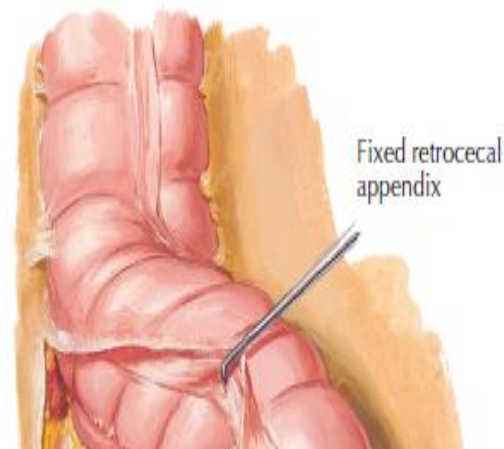
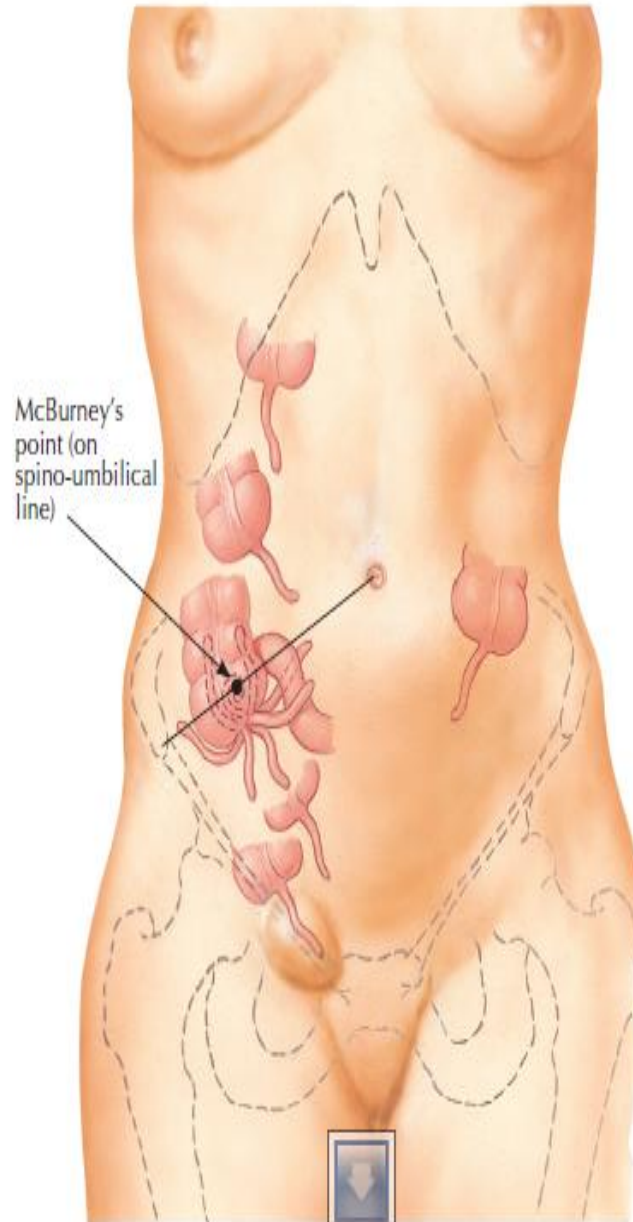
♣ 1.5 m long, surround the coils of small intestine

Consists of



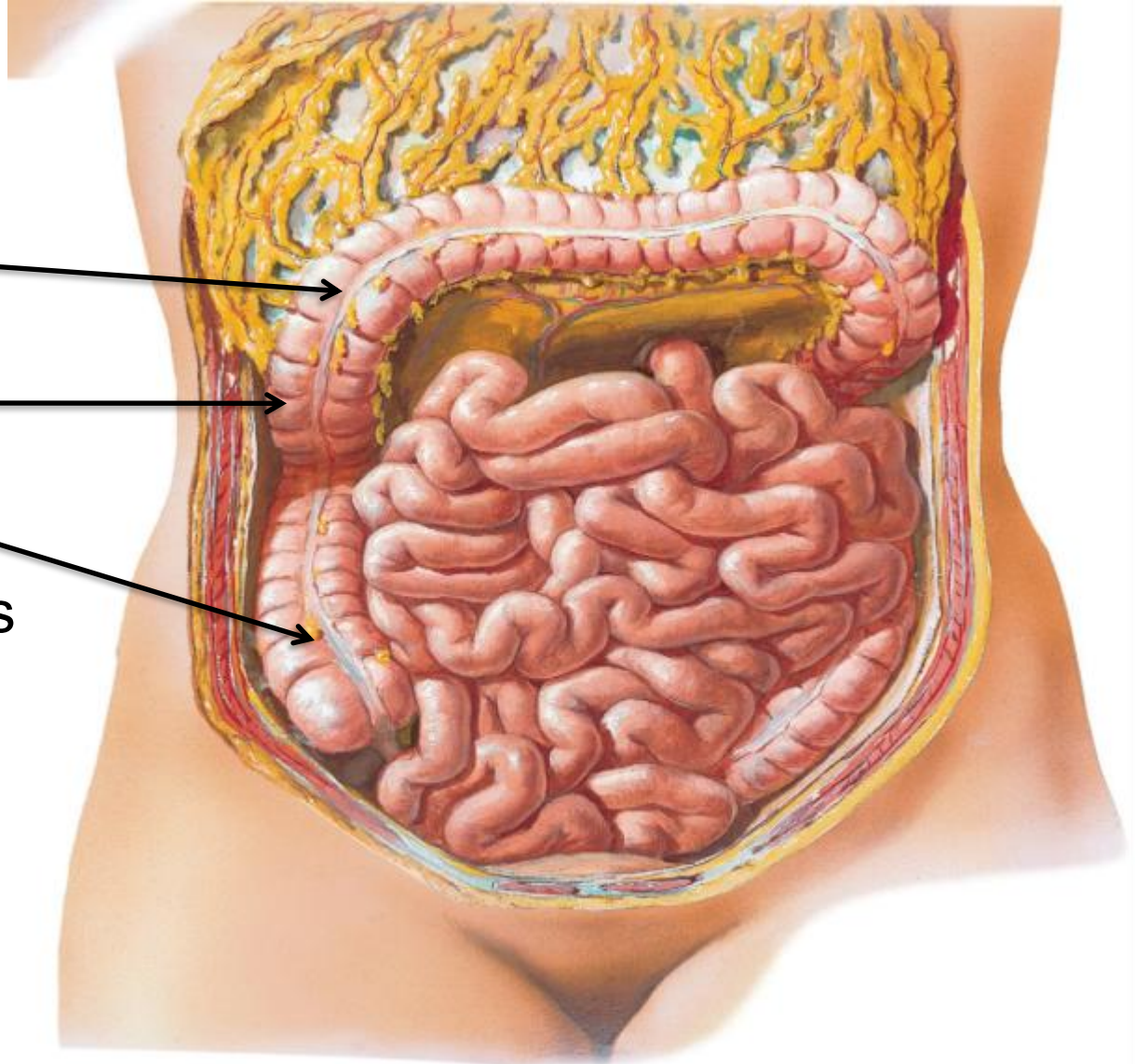
McBurney's point

The appendix lies in the right iliac fossa, and in relation to the anterior abdominal wall its base is situated one third of the way up the line joining the right anterior superior iliac spine to the umbilicus (McBurney's point)



Different between Small and large Intestine

- Taeniae coli
- Haustra: sacculations
- Omental appendices: small fat accumulations
- Greater luminal diameter



The second group of organs are the **accessory organs**. These contribute to the break down of food into usable components (digestion) but are not part of the alimentary tract. The ***accessory organs*** are:

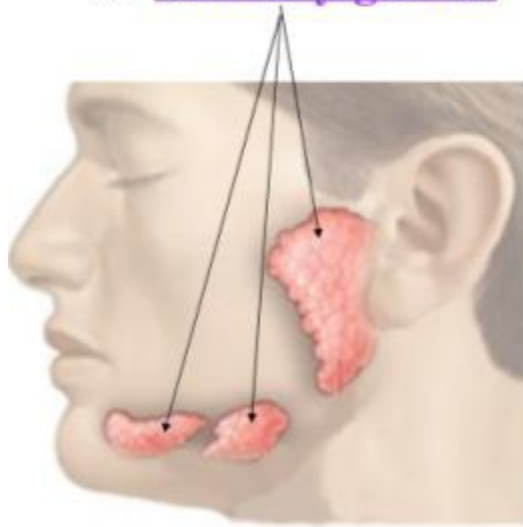
1. **Tongue**



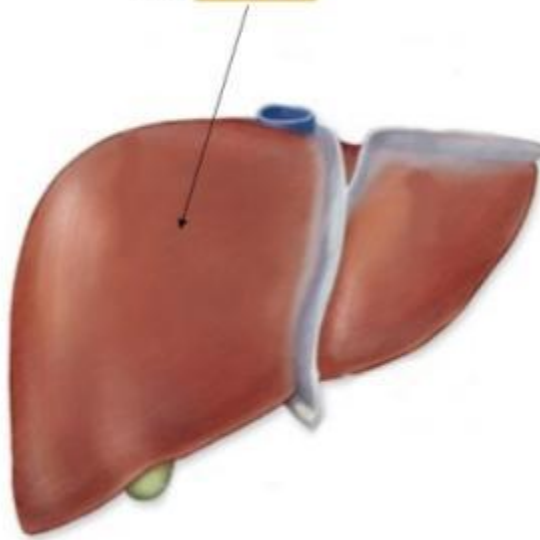
2. **Teeth**



3. Salivary glands



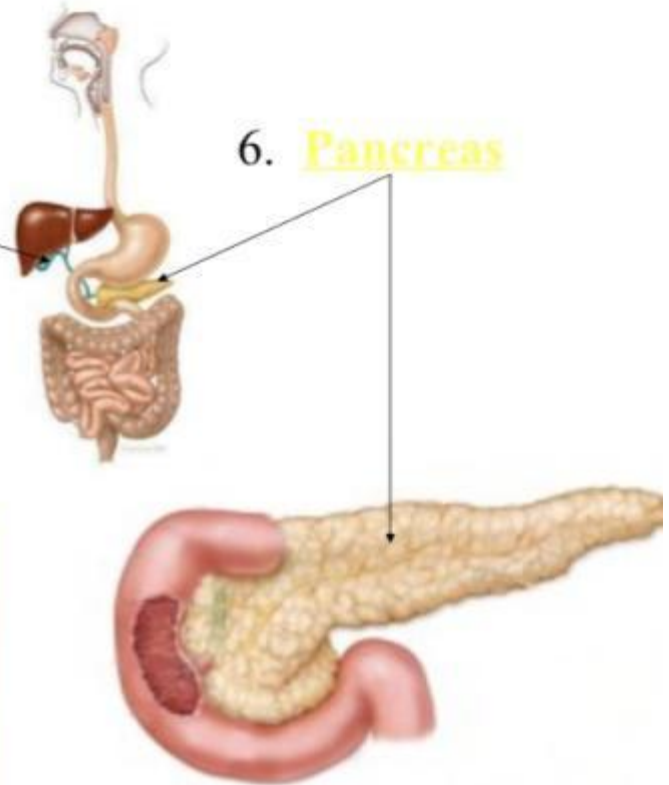
4. Liver



5. Gall bladder



6. Pancreas



Associated digestive Glands

1- The Salivary Glands

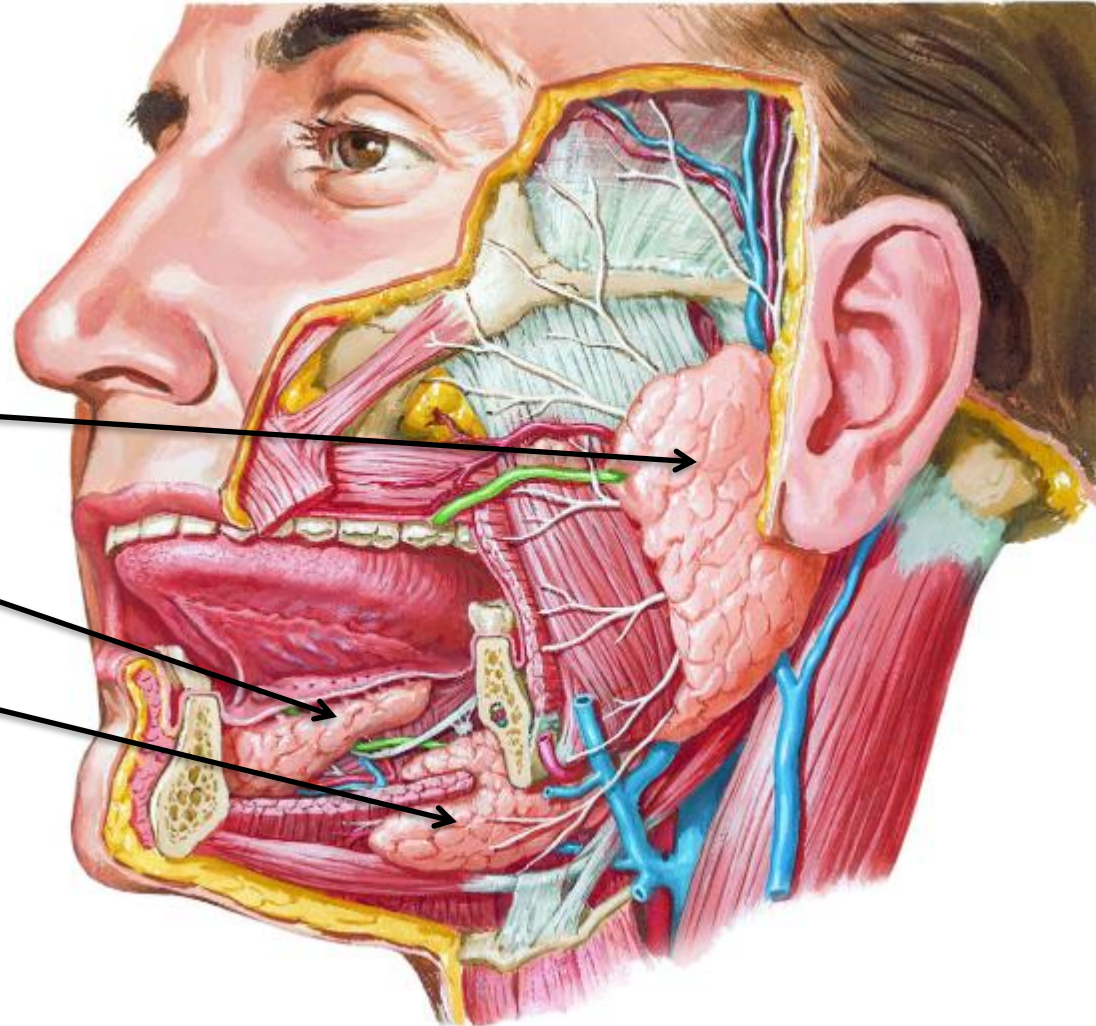
♣ Include three pairs;

Parotid

Sublingual and

Submandibular

♣ Their ducts open
in the oral cavity



Parotid gland

- The largest one located anterior and below the auricle on the ramus of mandible.
- Extends from external auditory meatus above to the upper part of the side of neck on the sternomastoid muscle.
- Has a fibrous capsule and a fascial sheath derived from the investing layer of deep fascia of neck.
- The **parotid duct (Stensen's duct)** emerges from the superficial part, crosses over masseter pierces buccinator muscle and overlying fat & fascia to open into mouth vestibule opposite the 2nd upper molar tooth

Submandibular Gland

- The 2nd largest salivary gland located in the neck, deep to the body of mandible (in the submandibular fossa).
- Has fibrous capsule and invested by a sheath of fascia from investing deep fascia of neck.
- The **submandibular duct (Wharton's duct)** emerges from the deep part to open in the floor of mouth cavity proper by a small papilla at the side of the frenulum of tongue.

Sublingual Gland

- The smallest pair of salivary glands located in the floor of mouth cavity under the tongue.
- A **major sublingual duct (Bartholin duct)** also joins the submandibular duct.

The Teeth

- The **deciduous teeth** of childhood are 20; 4 incisors, 2 canines, and 4 molars in each jaw. The first eruption starts at 6 months usually in lower jaw and are completed by the age of 2 years.
- The **permanent teeth** are 32: 4 incisors, 2 canines, 4 premolars and 6 molars in each jaw. The last molar teeth are usually erupted by the age 17-30.

The Tongue

- A mass of skeletal muscles covered by mucus membrane, located in oral cavity and oropharynx, and attached by its muscles to the styloid process of skull, mandible & hyoid bone.
- The right & left halves of tongue are connected together by a **median fibrous septum**.

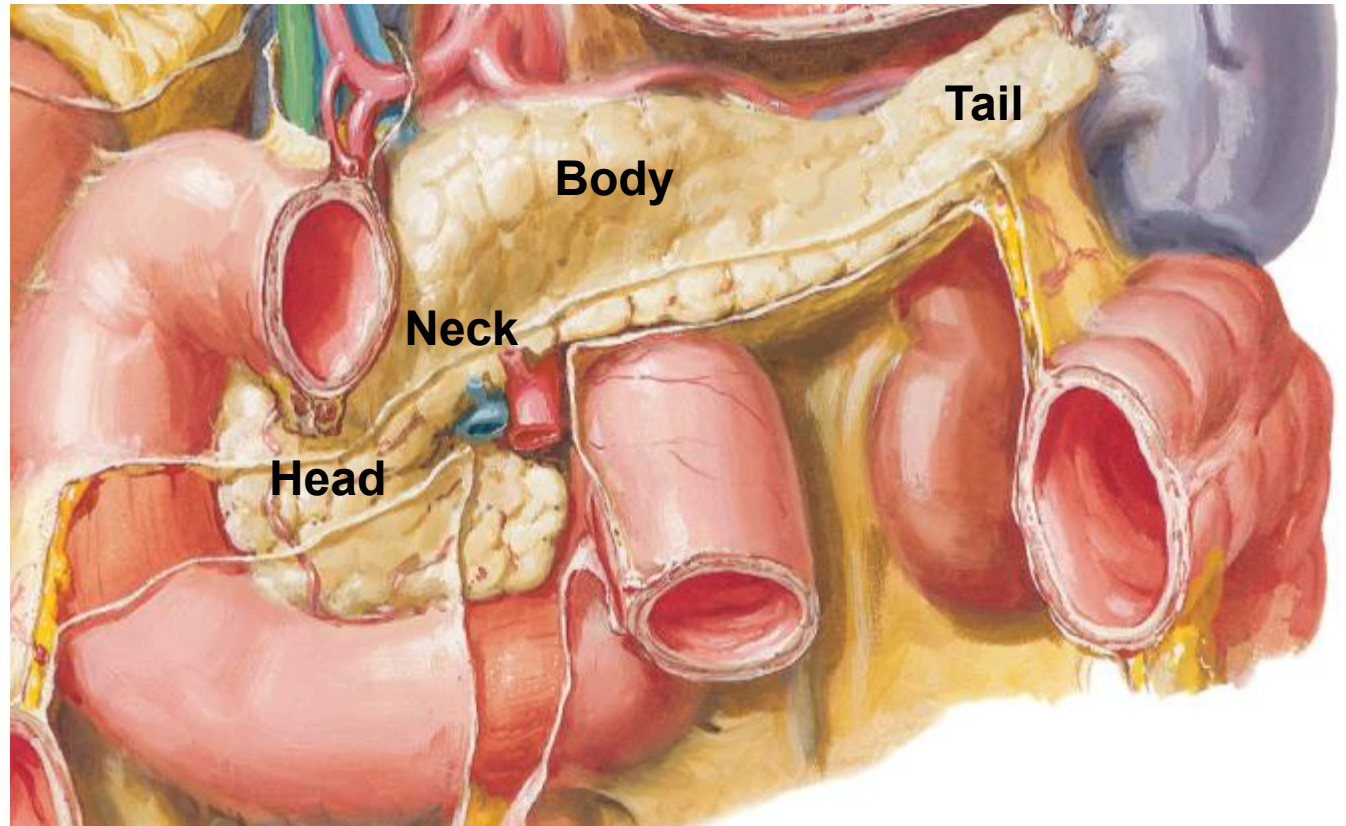
The Pancreas

- Is a soft, pink triangular gland located on the posterior abdominal wall reteroperitoneally.
- It produces digestive enzymes which are secreted into the duodenum.
- It is also an endocrine gland secreting insuline and glucagon.

The Pancreas

- ♣ Large gland surrounded by duodenum
- ♣ It is an endocrine and exocrine

Divided into
Head, Neck,
Body and Tail



Liver

- Liver is the largest gland in the body. It is located under the diaphragm, more to the right side of the body
- It has many metabolic functions.
- The digestive function of liver is by secretion of bile through the biliary ducts to the duodenum.
- The liver lobes are:
 - Right lobe
 - Left lobe
 - Caudate lobe
 - Quadrate lobe.

3- The Liver & Gall Bladder

♣ The liver is large organ

♣ Anteriorly right

and left lobes

By the Falciform lig

Visceral surface

The right is subdivided into

quadrate lobe

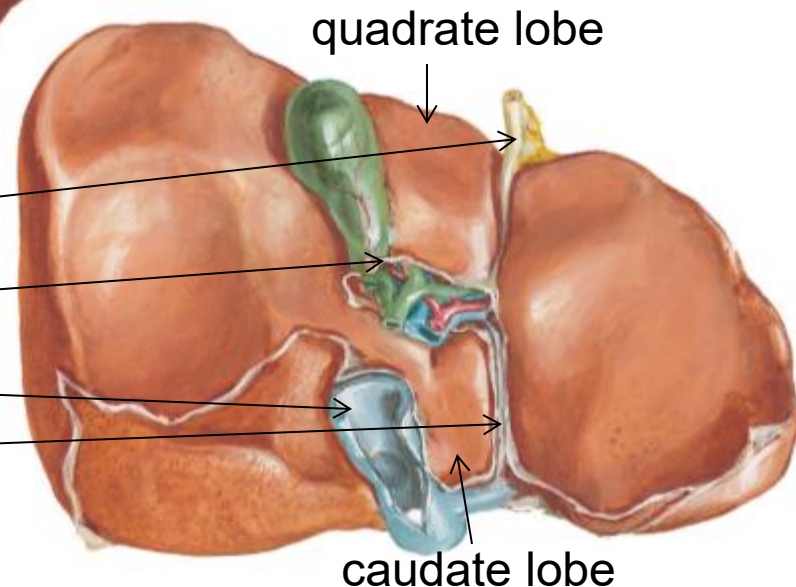
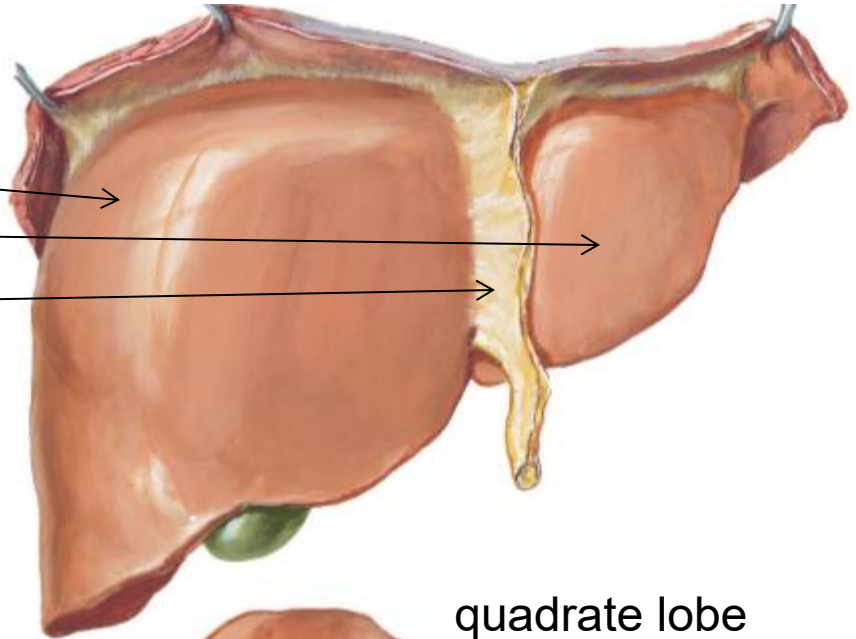
and a caudate lobe

by the ligamentum teres,

gallbladder

the inferior vena cava,

and ligamentum venosum

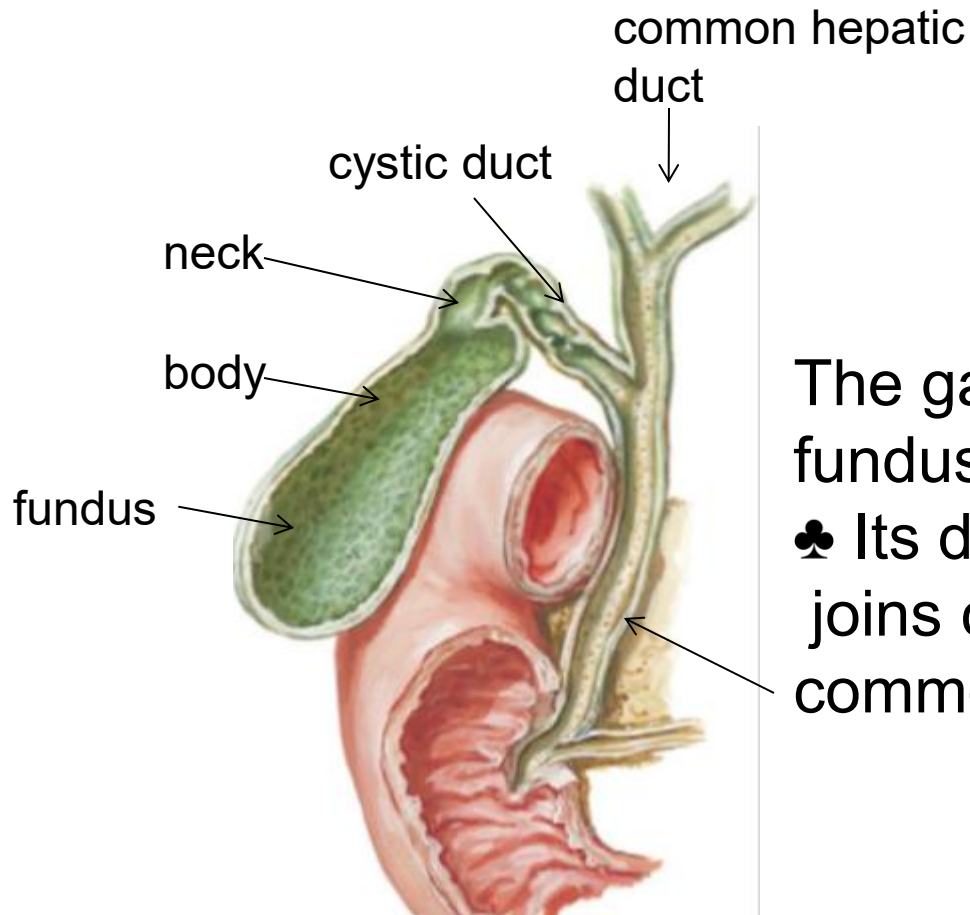
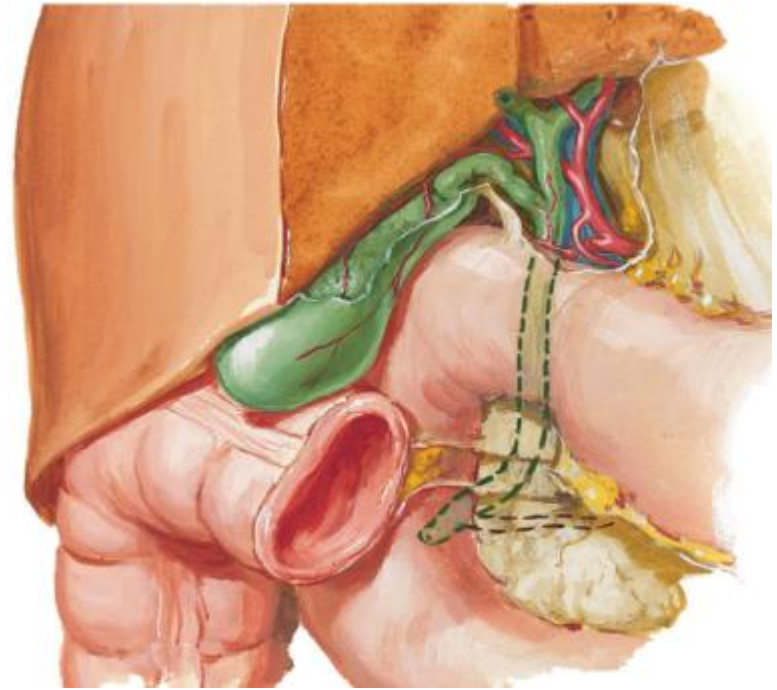


Gall bladder

- Gall bladder is a small thin walled green sac that snugles in a shallow fossa in the inferior surface of the liver.
- Right and left hepatic ducts unites to form the common hepatic duct which unites with the cystic duct to form the common bile duct.
- Common bile duct opens in the duodenum.

4- Gall bladder

♣ is muscular sac attached to inferior of visceral surface of liver

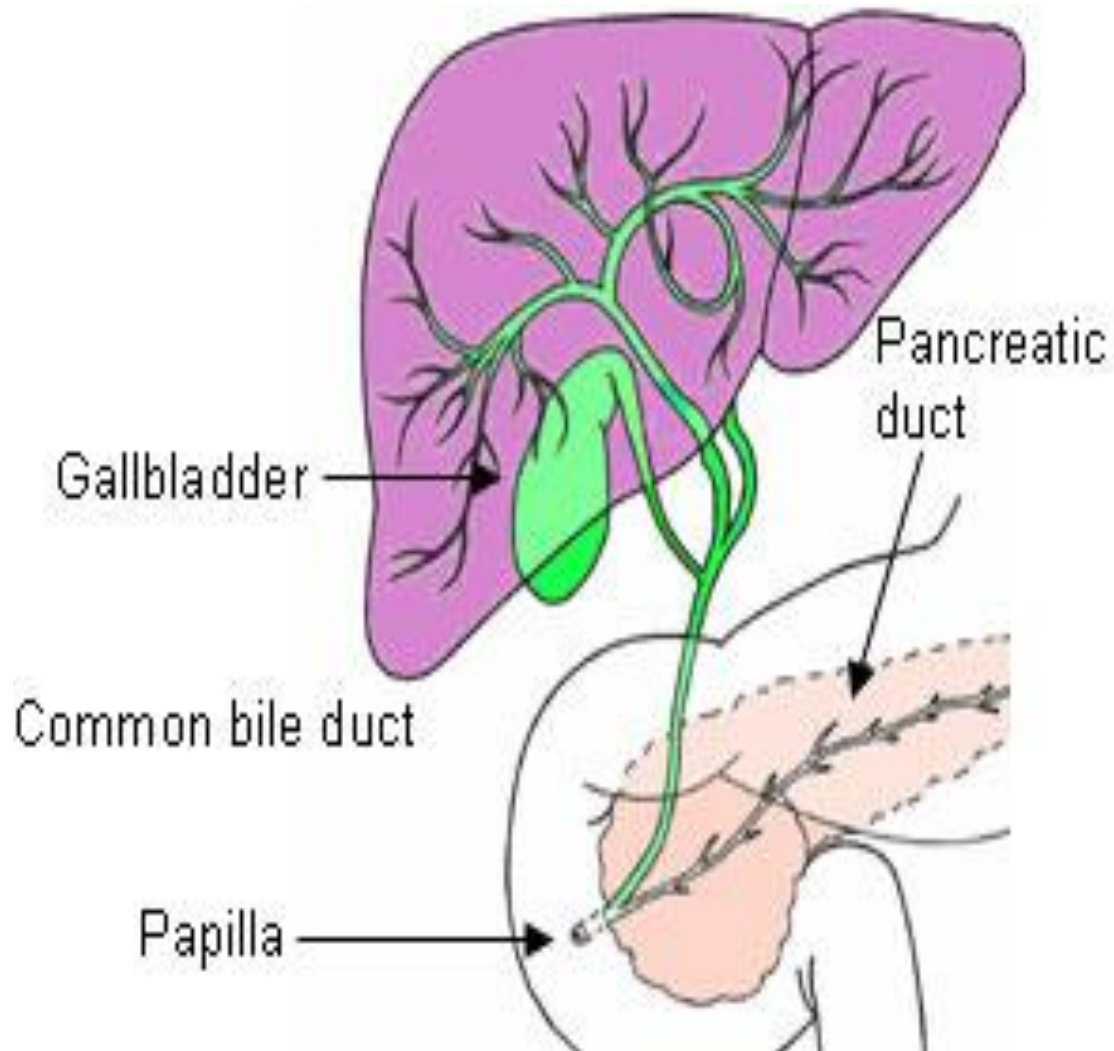


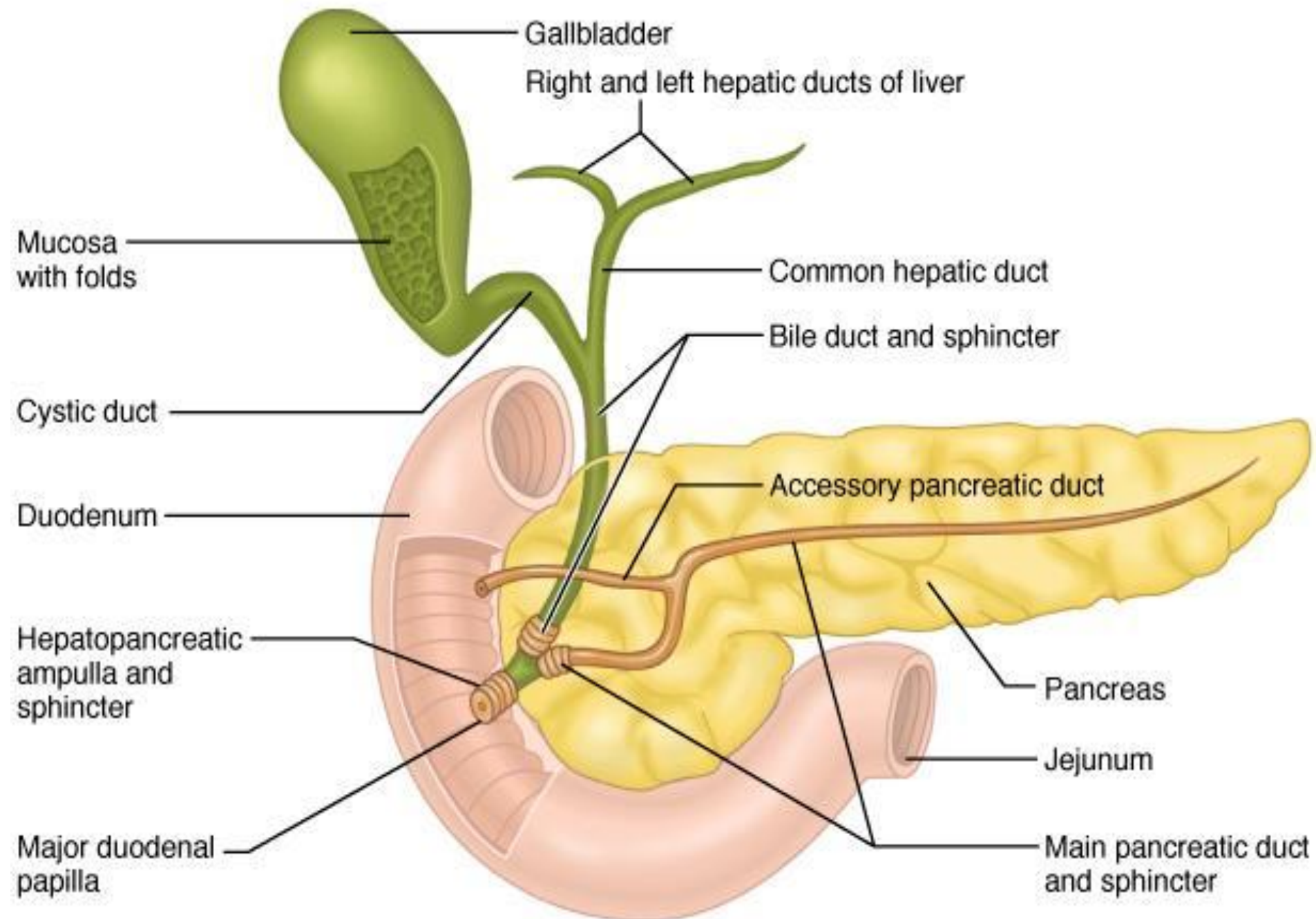
The gallbladder is divided into the fundus, body, and neck.

♣ Its duct is called cystic duct joins common bile duct to form common bile duct

The Biliary Passages

- ♣ Small ducts unite form Rt and Lt hepatic ducts
- ♣ They units and for common hepatic duct
- ♣ The cystic and CHD join together form CBD
- ♣ The CBD join with the pancreatic duct form Hepatopancreatic papilla





Peritoneum

- Is the largest serous membrane in the body.
- **Parietal peritoneum** is that portion of peritoneum that lines the walls of abdominopelvic cavity.
- **Visceral peritoneum** covers the organs.
- The slim space between visceral and parietal peritoneum is called **peritoneal cavity** which contains peritoneal fluid.
- **Ascitis** is accumulation of excess fluid inside the peritoneal cavity.

- Some organs lie on posterior abdominal wall and are covered only partially by peritoneum on their anterior surface they are called **retroperitoneal organs**.
- Some organs are surrounded by peritoneum and peritoneal cavity from all the sides called **intraperitoneal organs**.
- **Examples of retroperitoneal organs:**
 - Kidneys, suprarenal glands, ureters, pancreas, duodenum, ascending and descending colons,
- **Examples of intraperitoneal organs are:**
 - Stomach, spleen, liver, transverse colon, sigmoid colon, jejunum, ileum.

- There are five major peritoneal folds: the *greater omentum, falciform ligament, lesser omentum, mesentery, and mesocolon..and others.*

